

AYA MEKNI

AI Engineer · LLM Systems · Intelligent Agents · AI Automation

Monastir, Tunisia · +216 92 819 319 · aya.mekni@esprim.tn · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

Open to relocation · Remote-friendly

PROFILE

AI Engineer specializing in production-grade multi-agent systems, RAG architectures, and LLM deployment at scale. Proven ability to ship AI products end-to-end — from agent design and cost optimization to observability and multi-channel deployment. Thrive in cross-functional environments where AI complexity meets real business constraints. Open to full-time roles.

TECHNICAL SKILLS

LLM & Agents RAG, multi-agent orchestration, tool/function calling, prompt engineering, prompt caching, memory management, multi-LLM routing, LLM evaluation, observability, cost optimization, human-in-the-loop

Frameworks & Models LangChain, LangGraph, CrewAI, ReAct, MCP, OpenAI GPT, Anthropic Claude, LLaMA, Mistral, HuggingFace

ML & NLP Transformers, embeddings, semantic search, fine-tuning, NER, OCR, Document AI, PyTorch, LSTM, Pandas, NumPy

Cloud & AI Services Azure OpenAI, Azure AI Search, Azure ML, Azure Data Factory, Docker, GitHub Actions, CI/CD, Linux

Databases PostgreSQL, Neo4j, Pinecone, FAISS, ChromaDB, SQL, NoSQL

Backend & APIs Python, FastAPI, Django, Flask, REST APIs, WebSockets, Microservices, React/Next.js

MLOps & Data MLflow, Airflow, Spark, Kafka, CI/CD, experiment tracking, model deployment

Automation & Integration n8n, Power Automate, WhatsApp API, Telegram Bot API, Slack API, token cost optimization

EXPERIENCE

AI Engineer · Roundesk (Full-time · Hybrid · Sousse · Agile/Scrum · GitLab · Jira)

Nov 2025 – Present

- Architected a **multi-tenant agentic SaaS platform** using **LangGraph** and **ReAct**, with configurable LLM backends and inter-agent delegation across structured agent workflows
- Engineered **MCP servers** connecting external APIs (CRM, ticketing, internal tools) into agent tool ecosystems with permission-based execution controls per tenant
- Built a two-mode **escalation system** — deterministic trigger-word handoff and **confidence-based escalation** with auto-generated summaries — reducing unnecessary escalations by **~35%** vs. trigger-only baseline
- Layered a **persistent memory system** combining short-term context windows with long-term **vector-stored history**; applied **conversation summarization** cutting input tokens by **~50%** on sessions exceeding 10 turns
- Developed **multi-KB RAG fan-out** with **score-based reranking** across multiple knowledge bases before each LLM call
- Implemented **multi-LLM routing** and per-user billing logic across **OpenAI, Anthropic, Groq, and Google** via a unified provider abstraction layer
- Instrumented **per-turn observability logging** — decision traces, tool call outcomes, latency, and **hallucination signals** — across all production deployments
- Shipped the full stack via **Docker** with **CI/CD pipelines** and documented **FastAPI** microservices in weekly Agile sprints

AI Engineer Intern · CarthaPlay (Part-time · Remote)

Jun – Sep 2025

- Built a hybrid **LSTM + Transformer** behavior analysis pipeline over **7,000+ gameplay logs** — detecting temporal learning patterns (response time, error rates, guessing behavior) with a **20% lift** over rule-based baseline
- Developed a **RAG system (LangChain + Neo4j + LLM)** generating explainable, evidence-backed student recommendations with per-recommendation **confidence scores** and interpretable justifications delivered to educator dashboards
- Deployed the full inference pipeline via **FastAPI**, meeting the **<10s response SLA** under concurrent load
- Engineered **ETL pipelines** for gameplay log ingestion, cleaning, and aggregation feeding both the detection models and the RAG knowledge base

- AI Engineer Intern · ESPRIM — AffriOffre Project (Full-time · On-site)Jun – Jul 2025
- Led AI engineering in a 2-person team to build a **pan-African tender aggregation platform** — aggregating **500+ public tenders** across 5 countries via a hybrid scraping pipeline (**BeautifulSoup + Playwright**) with **>95% success rate**, proxy rotation, and scheduled runs
 - Built a **semantic search and recommendation engine** combining **TF-IDF, dense embeddings, and Azure AI Search** — serving multilingual filtered results at **<300ms API response time**
 - Developed a **RAG assistant** for bid preparation (**LangChain + LLM**) enabling users to chat with tender documents, extract key requirements, and generate structured proposal outlines
 - Integrated **multi-channel notifications** (email, **WhatsApp Business API**, WebSocket dashboard) and a platform chatbot backed by a **vector knowledge base (ChromaDB + RAG)**

- Software Engineer Intern · Anypli (Full-time · On-site · Agile/Scrum)Jun – Aug 2024
- Built **Eventify**, a real-time event management platform (**Laravel, WebSockets, Power BI**) — delivering a **25% increase in user engagement** through live interaction features
 - Reduced **SQL query latency by 35%** via **Eloquent ORM** tuning and query caching; implemented **MFA and RBAC** with OWASP-compliant session security
 - Participated in Agile/Scrum ceremonies, code reviews, and technical documentation within a cross-functional production engineering team

- Freelance AI & Full-Stack Developer · Independent2023 – 2025
- Designed and developed full-stack web applications for **5+ small business clients** across e-commerce and services sectors — covering **UI/UX design, CRUD development**, and production deployment using **React/Next.js, Django, Flask**
 - Built and deployed **conversational AI assistants (LangChain + Mistral/LLaMA/OpenAI)** across Telegram and web channels — handling customer support, FAQ automation, and lead qualification for **3+ clients**
 - Integrated **matching and recommendation systems** into **2 marketplace platforms**, improving product discovery and user retention
 - Engaged as AI specialist by development teams — owning the full **AI/ML layer** across **4+ client projects** from architecture to deployment

AWARDS

Winner — **AI Camera Challenge Hackathon** · Global AI Conference (GAICA) 2025
Real-time computer vision system with automated alerting pipeline on NVIDIA GPU infrastructure

CERTIFICATIONS

NVIDIA (2025) Building RAG Agents with LLMs · Transformer-Based NLP Applications · Fundamentals of Deep Learning · Generative AI with Diffusion Models · AI for Predictive Maintenance

365 Data Science (2025) LLM Engineering in Practice · Conversational AI Memory with LangGraph · Vector Databases with Pinecone

DataCamp (2025) RAG with LangChain · Working with Llama 3 · Introduction to MLflow

KodeKloud (2025) Docker · Fundamentals of DevOps · Shell Scripts · AI-900 Azure AI Fundamentals

EDUCATION

National Engineering Diploma — Computer Science (Data Science & AI)Graduated 2026
ESPRIMGPA: 15.12 / 20

LANGUAGES

Arabic Native | English C1 | French C1 | German A2 (currently learning)